A SUBRING OF AN ALGEBRAIC FIELD EXTENSION CONTAINING THE BASE FIELD IS A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The lemma is the reason intermediate rings of algebraic extensions never appear as proper subrings; it is used silently across the Galois correspondence sections.

Editorial note. The label lemma-subalgebra-algebraic-extension-field names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Same chapter: lemma-finite-is-algebraic (delivered card L48360: monogenic subextensions of algebraic extensions are finite), so the powers of a nonzero element are linearly dependent over the base

Source reference. Original reference 09GG: sources/stacks-fields/pinned-original-source.tex, source lines 878–878. Prerequisite pointer invoked at 1010; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/F be a finite or more generally an algebraic extension of fields. Any subring $F \subset R \subset E$ is a field.*

Proof. Let $\alpha \in R$ be nonzero. Then $1, \alpha, \alpha^2, \ldots$ are contained in R. By Lemma 09GG we find a nontrivial relation $a_0 + a_1\alpha + \ldots + a_d\alpha^d = 0$ with $a_i \in F$. We may assume $a_0 \neq 0$ because if not we can divide the relation by α to decrease d. Then we see that

$$a_0 = \alpha(-a_1 - \ldots - a_d\alpha^{d-1})$$

which proves that the inverse of α is the element $a_0^{-1}(-a_1 - \ldots - a_d\alpha^{d-1})$ of R. □

Date: stacks-project:fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.